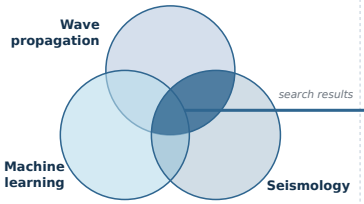
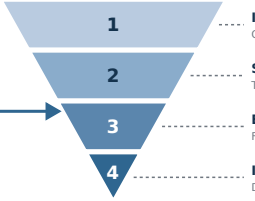
# A Scoping Review of Physics-Informed Machine Learning for Wave Propagation Modeling in Seismology

## 1 Literature search




## 2 Scoping review steps



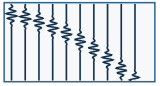

**Identification**
Combined keyword search in databases

**Screening**
Titles and abstracts for criteria

**Eligibility**
Full-text assessment of studies

**Inclusion**
Data charting and thematic synthesis

## 3 Trends



Seismic records



PIML



Inferred velocity model

**Recurrent approaches in seismology**

| | |
|---|---|
| Fourier neural operator | Generative adversarial |
| Recurrent network | Dual networks |

## 4 Challenges and oportunities

### Comparison

Accuracy and efficiency must be benchmaked against standard methods

### Evaluation

Must consider the full workflow: data generation, training, evaluation, and reuse across problems.

### Scalability

3D, high-frequency and field-scale validation remains an open challenge.